

TPT Agent

Algorithm Documentation

Tradier Paper Trading — Full Strategy Reference

Strategy	Cash-Secured Puts (CSP) + LEAPS Deep ITM Calls
Broker	Tradier Sandbox (Paper Trading) — Account VA54665450
Schedule	Daily at 6:35 AM PT (every day)
Data	Tradier real-time APIs + yfinance (earnings only)
Discord	#beta-ai-trades — Summary cards only
Version	Generated September 06, 2026

1. Overview

The TPT Agent is a fully automated options trading bot that runs on Tradier's paper trading sandbox. It executes two parallel options strategies on the same curated stock watchlist: Cash-Secured Puts (CSP Wheel strategy) and LEAPS deep ITM calls (stock replacement). The bot runs once per day at 6:35 AM PT, screens the entire approved stock universe, manages existing positions, and opens new ones based on a rules-based scoring system.

Key design principles: (1) Never trade below the 200-day SMA. (2) Never sell a CSP into an earnings window. (3) Size up only on high-conviction setups (score 5/5). (4) Close winners early — CSPs at 50% premium captured, LEAPS at +5% profit. (5) VIX gates deployment amount; LEAPS gate is VIX > 15.

System Architecture

Component	Detail
Bot file	tpt_agent.py
Scheduler	macOS launchd — com.ankushsinghal.tptagent.plist
Trigger time	6:35 AM PT, every day
Broker	Tradier sandbox (paper trading)
Account	VA54665450
Stock universe	Google Sheets CSV — curated approved watchlist
Options data	Tradier markets/options/chains (greeks=true) — real-time
Stock price data	Tradier markets/history + markets/quotes — real-time
VIX	Tradier markets/quotes (\$VIX.X) — yfinance fallback
Earnings filter	yfinance only (Tradier has no earnings calendar)
Order type	Limit orders at live mid-price (bid+ask)/2
Log file	tpt_agent.log
Discord channel	#beta-ai-trades (hardcoded webhook — no other channel)
Discord output	Run summary + open positions table + CSP summary + LEAPS summary

2. Data Sources

All market data is fetched in real-time via Tradier's REST APIs. yfinance is retained only for earnings dates, which Tradier does not provide.

Data Point	Source	Latency	Used For
Stock OHLCV bars	Tradier /markets/history	Real-time	SMA 20/50/200, Bollinger Bands, RSI
Real-time stock quote	Tradier /markets/quotes	Real-time	Current price, day change %
Options chain (bid/ask/OI)	Tradier /markets/options/chains	Real-time	Strike selection, ARR, OI filter
Implied Volatility (IV)	Tradier greeks.mid_iv	Real-time	IV scoring criterion, BS delta fallback
Options Delta (CSP)	Tradier greeks.delta	Real-time	Delta filter vs BB-position-based range (per stock)
Options Delta (LEAPS)	Black-Scholes computed	Real-time	Always recomputed — Tradier greeks unreliable for 1-2yr options
VIX	Tradier \$VIX.X → yfinance fallback	Real-time	Deploy % and LEAPS gate
Earnings dates	yfinance (only yfinance use)	Daily	Hard filter — skip if earnings within EARNINGS_FILTER_DAYS (10)
Option expirations	Tradier /markets/options/expirations	Real-time	Finding valid DTE windows
Live mid-price at execution	Tradier /markets/quotes	Real-time	Limit price on every order placed

3. Run Pipeline — 9 Phases

Every time the bot fires at 6:35 AM PT it executes nine sequential phases. Each phase depends on the results of the previous one.

1

VIX + Account Snapshot

Fetch VIX, portfolio value, cash, and all open positions from Tradier.

2

Manage Existing Positions

Profit exits placed automatically. Loss exits post Discord alert — trader closes manually.

3

Refresh Capital After Closes

Re-fetch account after any closes to get updated cash and buying power.

4

Load Tickers + Hard Filters

Pull approved stock list, fetch price history, apply 200 SMA and earnings hard filters.

5

CSP Screening + Scoring

Score each stock on 5 technical signals, find best put contract, rank by score then ARR/Delta ratio.

6

LEAPS Screening

If VIX > 15: keep names near the lower BB (dip), sort by proximity, find deep ITM call.

7

Execute LEAPS Trades

Top-ranked first, 1 contract per pick, until 15% portfolio budget exhausted.

8

Execute CSP Trades

Top-scored first, 1 contract per pick, until VIX-adjusted cash budget exhausted.

9

Post Discord Summary

Post run summary, open positions table, CSP summary table, LEAPS summary table.

4. Phase 1 — VIX & Account State

CSP Delta — BB-Based Per Stock (replaces VIX-global range)

CSP put delta is no longer determined by VIX globally. Each stock gets its own delta range based on its Bollinger Band position at screening time. See Phase 5 (Section 7) for the full 3-tier table.

VIX → Cash Deployment Percentage

VIX Level	Deploy %	Rationale
> 25	80%	Fear = opportunity, deploy aggressively
20 – 25	70%	Elevated vol, still good entry conditions
15 – 20	60%	Moderate vol, standard deployment
12 – 15	40%	Low vol, be selective and conservative
< 12	20%	Very low vol, premiums too thin to deploy much

Budget Calculations

- **CSP Budget** = Cash × VIX deploy %
- **LEAPS Budget** = Portfolio Value × 15% – Current LEAPS Exposure (never exceeds 15% of portfolio total)
- **LEAPS VIX Gate** = Screen and execute LEAPS when **VIX > 15**

Rationale for LEAPS VIX gate (> 15): below 15 the market is too complacent — stocks haven't pulled back, so the dip-buy thesis has nothing to act on. Above 15 there is enough volatility for quality names to reach the lower Bollinger Band.

5. Phase 2 — Position Management

Before opening any new positions, the bot scans every open position. Three independent rules evaluate each position. **Profit exits are placed automatically. Loss exits post a Discord alert — the trader decides manually.**

- **Auto-order:** Profit exits only — bot places the closing order immediately.
- **Discord alert:** Loss exits only — bot posts a recommendation, no order placed. Trader reviews and closes manually.

CSP Exit Rules — Short Puts (checked in priority order)

Rule	Trigger	P&L; State	Action
#1 — 50% Capture	$(\text{Avg Entry} - \text{Current}) / \text{Avg Entry} \geq 50\%$	Profit	■ AUTO — BUY TO CLOSE
#2 — 21-DTE Exit (flat or profit)	$\text{DTE} \leq 21$ AND $\text{captured} \geq 0\%$	Profit / Flat	■ AUTO — BUY TO CLOSE
#2 — 21-DTE Exit (small loss)	$\text{DTE} \leq 21$ AND $-50\% \leq \text{captured} < 0\%$	Small loss	■■ ALERT — manual close
#3 — Stop-Loss	$\text{Current} \geq \text{Avg Entry} \times 3.0$ (loss = $2 \times$ premium received)	Loss	■■ ALERT — manual close

LEAPS Exit Rules — Long Calls

Rule	Trigger	P&L; State	Action
#1 — Profit Target (+5%)	GTC SELL TO CLOSE at entry $\times 1.05$, rested at open	Profit	■ GTC order fills on the book (Phase 2 auto-close = backstop)
#2 — Stop-Loss	$(\text{Current} - \text{Avg Entry}) / \text{Avg Entry} \leq -50\%$	Loss	■■ ALERT — manual close

Discord Loss Alert Format

When a loss exit triggers, the bot posts an alert to **#beta-ai-trades** containing:

- Position (compact format: e.g. NVDA \$195P Jun26)
- Entry price → current price
- Unrealized P&L (\$ and %)
- Trigger reason (which rule fired)
- Suggested action (BUY/SELL TO CLOSE N contracts at market)
- Confirmation: *No order placed — awaiting your decision.*

Order Execution Mechanics (profit exits only)

- Fetch live bid/ask from Tradier **markets/quotes** endpoint
- Calculate mid-price: **(bid + ask) / 2**
- Place a **limit order at mid-price**, time-in-force = day
- If no live quote available: fallback to last known price

6. Phase 4 — Stock Universe & Hard Filters

The bot loads a curated watchlist from Google Sheets, fetches 310 calendar days of daily OHLCV bars from Tradier for each ticker, computes technical indicators, then applies two hard filters. Any stock failing either filter is completely excluded from both CSP and LEAPS screening.

Technical Indicators Computed

Indicator	Calculation	Used In
SMA 20	Mean of last 20 daily closes	Bollinger Band midline, CSP scoring
SMA 50	Mean of last 50 daily closes	CSP scoring (strike above 50 SMA)
SMA 200	Mean of last 200 daily closes	Hard filter — must be above this
Bollinger Upper	$SMA\ 20 + 2 \times std(\text{last 20 closes})$	Reference
Bollinger Lower	$SMA\ 20 - 2 \times std(\text{last 20 closes})$	LEAPS scoring (near lower band)
RSI (14)	Wilder's smoothed 14-period RSI	CSP hard gate (≥ 65 =skip) + CSP scoring (< 50 =+1) + LEAPS scoring (< 70 =+1)
Day Change %	$(\text{current} - \text{prev_close}) / \text{prev_close}$	CSP scoring (healthy pullback)

Hard Filters

Filter	Logic	Skip if
200-day SMA	Is stock in long-term uptrend?	Price $\leq$ SMA 200
Earnings (CSP + LEAPS)	Earnings report imminent?	Earnings date falls within next EARNINGS_FILTER_DAYS (10) days
RSI overbought (CSP + LEAPS)	Is stock overbought?	RSI $\geq$ RSI_OVERBOUGHT_MAX (65)

■ Earnings data is fetched from yfinance — the only remaining yfinance dependency. Tradier does not provide an earnings calendar.

7. Phase 5 — CSP Screening & Scoring

Step 1 — RSI Hard Gate (shared filter)

RSI $\geq$ RSI_OVERBOUGHT_MAX (65) is filtered out in Phase 4 for **both** CSP and LEAPS — overbought stocks have high mean-reversion risk. By the time CSP scoring runs, every candidate already has RSI < 65.

Step 2 — Pre-Score Gate (skip chain fetch early)

Before fetching any option chain, compute the **stock-level pre-score** (max 3): price below BB midline, healthy pullback today, RSI < 50. The other 2 score points (strike > 50 SMA, IV $\geq$ 40%) are contract-level. Since MIN_SCORE_TO_TRADE = 3 and at most 2 contract points can be added, any stock with pre-score < 1 can never qualify → **skip the chain fetch**. Correctness-preserving: it cannot drop a stock that would have qualified.

Step 3 — BB-Based Delta Range

BB Position	Delta Range	Rationale
Price $\leq$ Lower BB + 3%	0.25 – 0.35 (Aggressive)	Stock oversold → high reversal probability → sell closer to ATM
Between Lower BB and Mid BB	0.15 – 0.25 (Moderate)	Partial pullback → standard entry quality
Price $\geq$ Mid BB (SMA 20)	0.08 – 0.15 (Conservative)	At/above average → more downside room → stay far OTM

Step 4 — Technical Scoring (max 5 points)

Signal	Points	Condition
Above 50-day SMA	+1	Stock price > SMA 50 (medium-term uptrend)
At/below 20-day SMA	+1	Stock price $\leq$ SMA 20 / BB midline (short-term pullback)
Healthy pullback today	+1	Day change between -0.5% and -5.0%
RSI < 50	+1	Actively oversold — best entry timing (replaces price<\$100)
IV $\geq$ 40%	+1	Elevated implied volatility — premium is rich

Require final score $\geq$ MIN_SCORE_TO_TRADE (3) to qualify. Score = 5 → flagged HIGH CONVICTION.

Step 5 — Find Best Put Contract (Tradier Options Chain)

For each expiration date between MIN_DTE (30) and MAX_DTE (45) days:

- Fetch full options chain from Tradier with **greeks=true**
- Filter puts: Open Interest $\geq$ 50 (rejects OI = 0), valid bid/ask quote
- Get delta from **Tradier greeks.delta**; fallback to Black-Scholes if null
- Apply BB-position-based delta filter (per-stock tier: aggressive / moderate / conservative)
- Calculate **ARR = (mid / strike) $\times$ (365 / DTE) $\times$ 100**

Strike Selection Logic (ARR band 40–70%):

Contracts within band ($40\% \leq \text{ARR} \leq 70\%$): Pick the highest ARR in band.

If ALL contracts are above the cap ($\text{ARR} > 70\%$): Step down to the lowest (safest) ARR above the cap.

If NO contract reaches the floor (all $\text{ARR} < 40\%$): Skip the stock entirely — MIN_ARR is a HARD floor, never breached. (This is why e.g. a 30%-ARR contract will not be selected.)

Step 6 — Rank and Select Top 5

- Sort qualifying CSPs by the following priority:
- **1. Score DESC** — technical setup quality (primary)
- **2. $\text{ARR} \div \text{Delta}$ DESC** — normalized risk-adjusted return (secondary)
- Rationale: a lower delta + higher ARR means the market pays you MORE for taking LESS
- directional risk. ARR/Delta correctly ranks e.g. delta=0.14 ARR=69% (ratio=493)
- above delta=0.20 ARR=47% (ratio=235). Pure ARR alone would invert this.
- **3. DTE DESC** — more time = more cushion for recovery
- **4. ARR DESC** — final tie-breaker only
- Take top 5

8. Phase 6 — LEAPS Screening

Currently DISABLED (LEAPS_ENABLED = false): the bot runs CSP only. The logic below applies when LEAPS is re-enabled.

VIX Hard Gate: LEAPS screening runs ONLY when **VIX > 15**. Below 15 the market is too calm for the dip-buy thesis, so the entire LEAPS phase is skipped.

Runs on the stock universe that passed the Phase 4 hard filters (price > 200 SMA, RSI < 65, no earnings ≤ 10d).

Dip-buy thesis (mean-reversion): LEAPS target quality names that have **pulled back** — not momentum names making new highs. There is NO scoring; a single dip filter qualifies candidates and prioritization is by pure sort.

LEAPS Hard Filters

Filter	Logic	Rationale
Above 200-day SMA	Price > SMA 200 (Phase 4)	Knife guard — long-term uptrend intact
RSI < 65 (shared)	RSI(14) < 65 (Phase 4)	Not overbought (applies to CSP too)
RSI < 40 (LEAPS)	RSI(14) < LEAPS_RSI_MAX (40)	Only genuinely oversold names qualify
No earnings ≤ 10d	Phase 4 earnings filter	Avoid imminent earnings gap
Near lower BB	price ≤ lower BB × 1.05 (within LEAPS_BB_LOWER_PCT = 5%)	The dip — buy quality on a pullback

Note: 'above 50 SMA' was removed — it conflicts with the near-lower-BB dip thesis (a stock at its lower band is rarely comfortably above its 50 SMA).

LEAPS Prioritization (pure sort — no scoring)

- **1. Distance above lower BB, ASC** — closest to the band = most oversold = best dip entry
- **2. Distance above 200 SMA, DESC** — tiebreaker: strongest long-term trend on the deepest discount
- Take top 5

LEAPS Contract Selection (Tradier Options Chain)

- Expiration window: **365 to 730 days (1–2 years out)**
- **Farthest expiry first:** fetch the longest-dated expiry in the window (max time value, lowest theta — ideal for stock replacement). Fall back to the next-farthest only if it yields no qualifying contract. ~1 chain/stock vs 6–10 before.
- Contract type: **Call options only**
- OI filter: skip only if confirmed OI > 0 but < 50 (OI = 0 means data unavailable, allow through)
- Price: (bid+ask)/2 → ask → bid → lastPrice (in priority order)
- Delta: use **Tradier greeks.delta** if available, otherwise compute via **Black-Scholes**
- For BS computation: use Tradier IV if IV ≥ 0.20, else default to **0.45** (LEAPS greeks are unreliable in data feeds)
- Delta filter: **0.70 ≤ delta ≤ 0.85** (deep in the money)
- Selection: within the chosen expiry, pick the contract whose delta is **closest to 0.77 (target)**

9. Phases 7 & 8 — Trade Execution

LEAPS Execution (Phase 7)

Budget: Portfolio Value $\times$ 15% (minus current LEAPS exposure already on the books)

Allocation: Top-ranked first. 1 contract per pick. Full remaining budget available for each pick.
Skip a pick only if the remaining budget is too small for its cost — move to the next pick.

For each LEAPS pick (in ranked order):

- If remaining budget $<$ cost_per_contract $\rightarrow$ skip, try next
- Fetch live mid-price from Tradier quotes
- Place **limit BUY TO OPEN** order at mid-price, qty = 1, time-in-force = day
- **Immediately rest a GTC take-profit:** a **SELL TO CLOSE** limit at **entry $\times$ 1.05 (+5%)**, duration = good-till-cancelled — so the profit target fills intraday without waiting for the next daily run
- Deduct cost from remaining budget and continue to next pick

CSP Execution (Phase 8)

Budget: Cash $\times$ VIX deploy %

Collateral: Each CSP requires Strike $\times$ \$100 in cash as collateral.

Allocation: Top-scored first. 1 contract per pick.

For each CSP pick (in score-ranked order):

- If remaining cash $<$ strike $\times$ \$100 $\rightarrow$ skip, try next
- Fetch live mid-price from Tradier quotes
- Place **limit SELL TO OPEN** order at mid-price, qty = 1, time-in-force = day
- Deduct collateral from remaining cash and continue to next pick

Order Format (Tradier API)

All orders are placed as form-encoded POST requests to Tradier:

Parameter	CSP Value	LEAPS Value
class	option	option
symbol	Underlying ticker	Underlying ticker
option_symbol	OCC put symbol	OCC call symbol
side	sell_to_open	buy_to_open
quantity	1	1
type	limit	limit
price	Live Tradier mid	Live Tradier mid
duration	day	day

10. Key Parameters Reference

All parameters are set via environment variables in the launchd plist (com.ankushsinghal.tptagent.plist). The defaults below are the current live values.

CSP Parameters

Parameter	Default	Description
MIN_DTE	30	Minimum days to expiration for CSP puts
MAX_DTE	45	Maximum days to expiration for CSP puts
EARNINGS_FILTER_DAYS	10	Earnings hard-filter look-ahead — skip stock if earnings within this many days (applies to BOTH CSP and LEAPS; decoupled from MAX_DTE)
MIN_ARR	40%	Minimum annualized return on risk to qualify
MAX_ARR	70%	ARR cap — step down to safer strike if exceeded
MIN_OPEN_INTEREST	50	Minimum open interest for liquidity (rejects OI = 0 — CSP needs real liquidity)
RSI_OVERBOUGHT_MAX	65	Shared hard filter — skip stock (CSP + LEAPS) if RSI $\geq$ this
CSP_DELTA_NEAR_LOWER_MIN/MAX	0.25/0.35	Delta range when price $\leq$ lower BB + 3% (aggressive)
CSP_DELTA_MID_ZONE_MIN/MAX	0.15/0.25	Delta range when price between lower and mid BB (moderate)
CSP_DELTA_ABOVE_MID_MIN/MAX	0.08/0.15	Delta range when price $\geq$ mid BB (conservative)
MIN_SCORE_TO_TRADE	3	Minimum score (out of 5) to qualify for execution
SIZE_UP_SCORE	5	Score threshold for HIGH CONVICTION label
SCORE_IV_MIN_PCT	40%	IV threshold for +1 scoring point
SCORE_RSI_OVERSOLD	50	RSI below this earns +1 score point (actively oversold)
SCORE_DOWN_TODAY_MIN_PCT	0.5%	Min pullback today to earn the pullback score point
SCORE_DOWN_TODAY_MAX_PCT	5.0%	Max pullback today (beyond this = potential problem)
CSP_CLOSE_PREMIUM_PCT	50%	Close CSP when this % of premium has been captured
CSP_DTE_EXIT	21	Close CSP at DTE $\leq$ this with positive/neutral P&L; (gamma gate)
CSP_STOP_LOSS_MULT	2.0×	Close CSP when loss $\geq$ 2× premium received

LEAPS Parameters

Parameter	Default	Description
LEAPS_ENABLED	false	Master switch — LEAPS OFF (CSP only). Set true to re-enable the LEAPS leg.
LEAPS_MIN_DTE	365 days	Minimum DTE for LEAPS contracts
LEAPS_MAX_DTE	730 days	Maximum DTE for LEAPS contracts (1–2 years)
LEAPS_MIN_DELTA	0.70	Minimum call delta (lowered from 0.80)

Parameter	Default	Description
LEAPS_MAX_DELTA	0.85	Maximum call delta (lowered from 0.99)
LEAPS_TARGET_DELTA	0.77	Target delta — pick contract closest to this (lowered from 0.85)
LEAPS_VIX_MIN	15	LEAPS enabled only when VIX > this
LEAPS_RSI_MAX	40	LEAPS-specific RSI ceiling — require RSI < this (stricter than shared 65)
LEAPS_BB_LOWER_PCT	5%	Dip filter — price must be within this % of the lower Bollinger Band
LEAPS_MIN_OI	50	Minimum OI (only enforced when OI > 0 from API)
LEAPS_MAX_PORTFOLIO_PCT	15%	Hard cap on total LEAPS exposure as % of portfolio (raised from 10%)
LEAPS_RSI_PERIOD	14	RSI calculation period (Wilder's smoothed)
LEAPS_CLOSE_PROFIT_PCT	5%	Close LEAPS when unrealized profit reaches this (lowered from 25%)
LEAPS_STOP_LOSS_PCT	50%	Close LEAPS when unrealized loss reaches this

Performance Parameters

Parameter	Default	Description
MAX_WORKERS	6	Thread-pool size for parallel per-ticker screening (Phases 4, 5, 6)

Performance: per-ticker screening runs in parallel; option expirations are cached per ticker; LEAPS fetch only the farthest expiry; CSP and LEAPS gate on stock-level score before any chain fetch. A full run takes ≈ 1 minute (down from ≈ 7 minutes).

11. Discord Output

All output goes exclusively to **#beta-ai-trades**. The webhook URL is hardcoded in `tpt_agent.py` — it cannot be redirected to another channel via configuration. Only summary tables are posted; no individual trade detail cards.

Post #	Content	Always Posted?	Format
1	Run Summary	Yes	Embed: balance, VIX, profit closes (auto-ordered), loss alerts (manual), new positions
2	Open Positions Table	If positions exist	Code block: position, qty, P&L; \$, P&L; %
3	Loss Alert (per position)	If loss exit triggered	Red embed: position, P&L;, trigger reason, suggested action. No order placed.
4	LEAPS Criteria Embed	Yes	Embed: VIX gate status, criteria list, DTE/delta targets
5	LEAPS Summary Table	If VIX in range + picks found	Code block: ticker, strike, delta, expiry, cost
6	CSP Summary Table	If picks found	Code block: ticker, strike, delta, expiry, premium, ARR

All embeds and tables include `[TPT]` label to distinguish from the Experiment Bot which posts to the same channel.

12. Full Decision Flow

START 6:35 AM PT trigger

Fetch VIX → determine deploy %; LEAPS gate: $VIX > 15$

Fetch Tradier account: portfolio value, cash, option buying power

Fetch all open positions

■■■ POSITION MANAGEMENT ■■■

For each open SHORT PUT (CSP):

- If stop-loss triggered ($\text{loss} \geq 2 \times \text{premium}$) → ■■ DISCORD ALERT only, no order
- If 50% premium captured → ■ BUY TO CLOSE at mid (auto)
- If $\text{DTE} \leq 21$ and in profit → ■ BUY TO CLOSE at mid (auto)
- If $\text{DTE} \leq 21$ and at loss → ■■ DISCORD ALERT only, no order

For each open LONG CALL (LEAPS):

- If stop-loss triggered ($\text{loss} \geq 50\%$) → ■■ DISCORD ALERT only, no order
- +5% take-profit rests as a GTC SELL TO CLOSE from entry; Phase 2 auto-close is a backstop

Re-fetch account to capture released capital

■■■ STOCK SCREENING ■■■

Load approved ticker list from Google Sheets

For each ticker (parallel, ThreadPoolExecutor — $\text{MAX_WORKERS}=6$):

- Fetch 310 days OHLCV + quote from Tradier → compute SMA20/50/200, BB, RSI
- HARD FILTER 1: skip if $\text{price} \leq \text{SMA } 200$
- HARD FILTER 2: skip if earnings within $\text{EARNINGS_FILTER_DAYS}=10$ (yfinance)
- HARD FILTER 3: skip if $\text{RSI} \geq 65$ (overbought — both CSP & LEAPS)

■■■ CSP SCORING (parallel) ■■■

For each stock passing hard filters:

- Pre-score gate: skip chain fetch if stock-level pre-score can't reach 3
- BB-based delta range: near lower $\text{BB}=0.25\text{--}0.35$ / between $\text{BBs}=0.15\text{--}0.25$ / above mid $=0.08\text{--}0.15$
- Fetch put chain from Tradier (greeks=true)
- Filter puts: $\text{OI} \geq 50$ (rejects $\text{OI}=0$), delta in BB-tier range, ARR 40–70%
- Score: strike > 50 SMA (+1), below mid BB (+1), pullback 0.5–5% (+1), $\text{RSI} < 50$ (+1), $\text{IV} \geq 40\%$ (+1)
- If final score < 3 : skip

Sort CSPs: score DESC → ARR/Delta DESC → DTE DESC → ARR DESC → take top 5

■■■ LEAPS SCREENING (parallel; only if LEAPS_ENABLED and $VIX > 15$) ■■■

For each stock passing hard filters:

- RSI FILTER: skip unless $\text{RSI} < 40$ (LEAPS_RSI_MAX) — genuinely oversold
- DIP FILTER: skip unless price within 5% of lower BB ($\text{LEAPS_BB_LOWER_PCT}$)
- No scoring — survivors are all valid dip candidates
- Fetch ONLY the farthest expiry chain (DTE 365–730); fall back if no contract
- Filter calls: $\text{OI} \geq 50$, delta 0.70–0.85; pick closest to delta 0.77

Sort LEAPS: dist-above-lower-BB ASC → dist-above-200-SMA DESC → take top 5

■■■ EXECUTION ■■■
LEAPS: iterate top-ranked first, 1 contract each, until 15% budget exhausted; rest GTC +5% sell
CSP: iterate top-scored first, 1 contract each, until VIX-deploy cash exhausted
All orders: limit at live Tradier mid-price, time-in-force = day
■■■ DISCORD ■■■
Post run summary, open positions, CSP table, LEAPS table to #beta-ai-trades
END

** The experiment_bot.py (Alpaca) runs an identical strategy independently. Trade picks are generated separately for each broker — they are not shared. This allows side-by-side comparison of broker execution quality.*